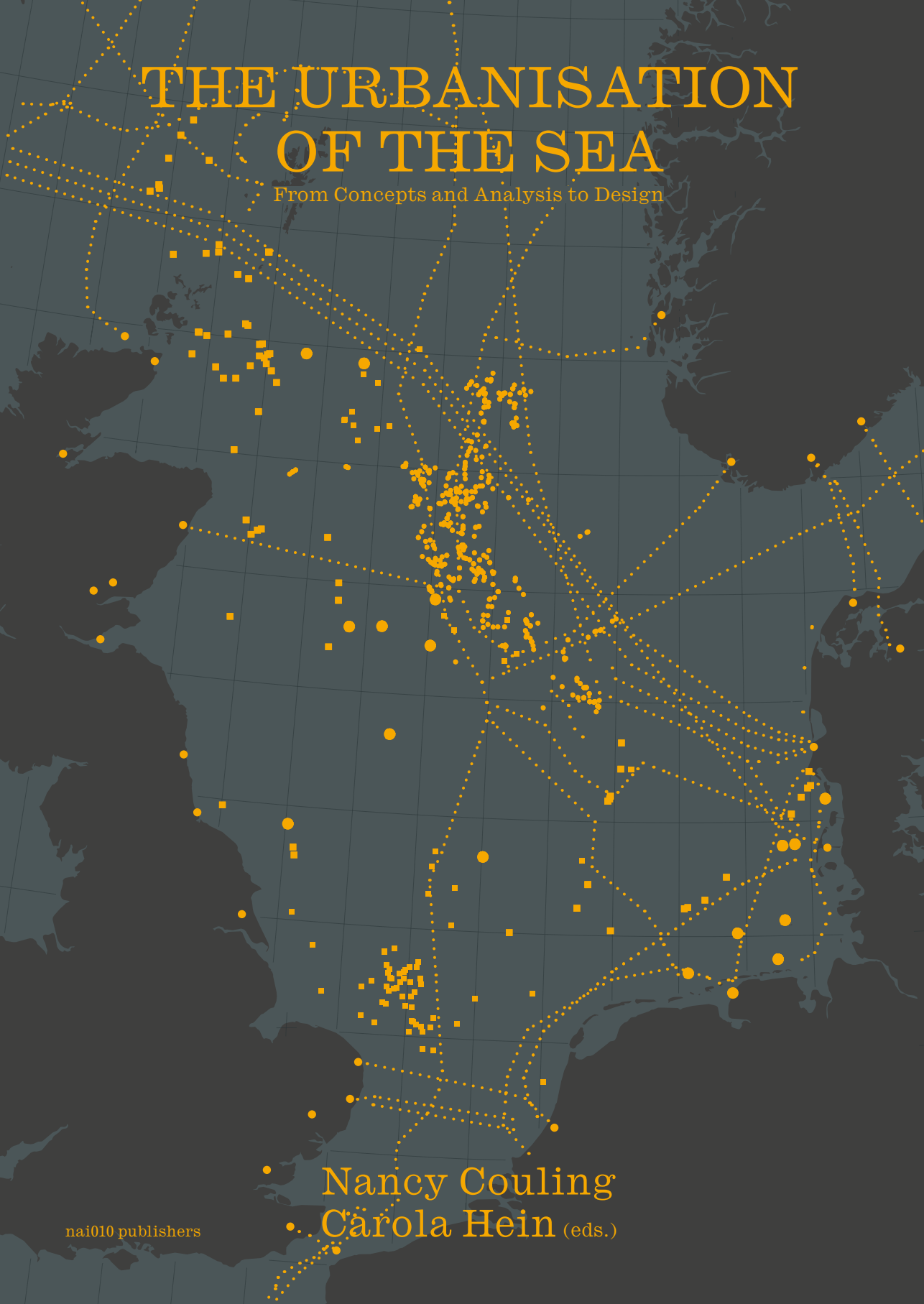




THE URBANISATION OF THE SEA

From Concepts and Analysis to Design

• Nancy Couling

• Carola Hein (eds.)

nai010 publishers

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Fig. 4
Below the Water
Towers (Design
Earth, Pacific
Aquarium 2016)

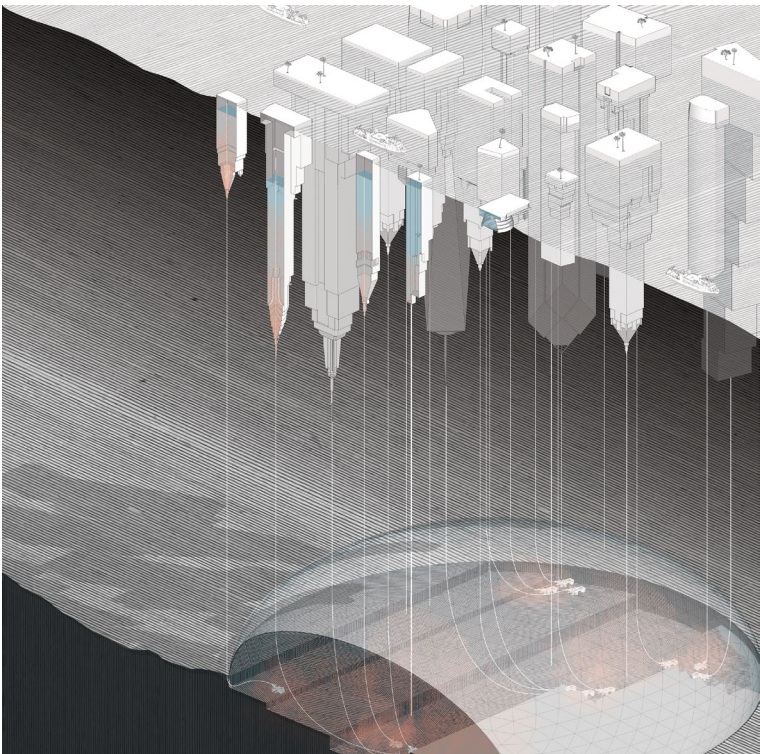
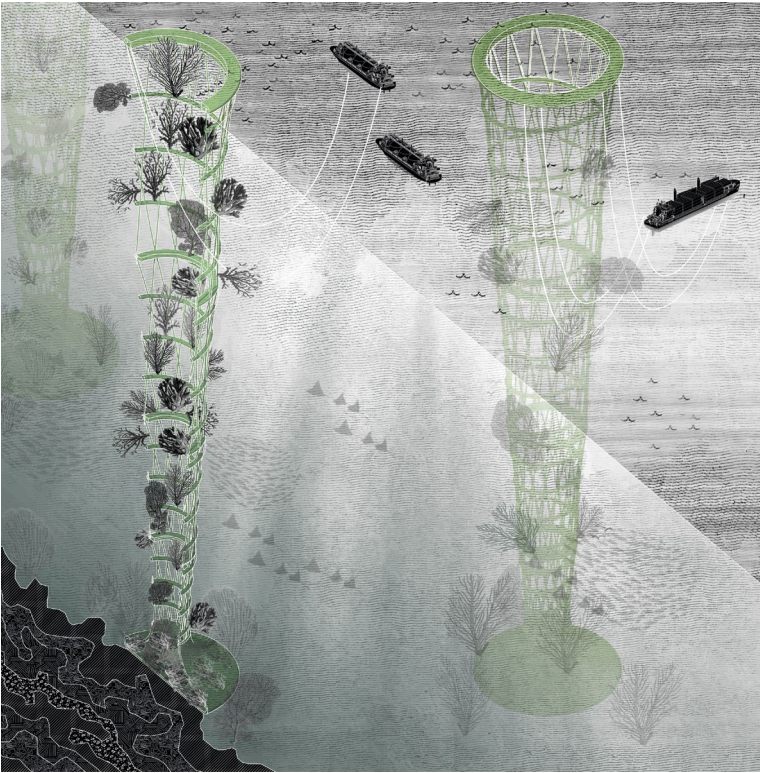


Fig. 5
Iron Towers
(Design Earth,
Pacific Aquarium
2016)



Chapter 20 AQUARIUM FOR THE ANTHROPOCENE*

Rania Ghosn & El Hadi Jazairy (DESIGN EARTH)

The ocean is inextricably linked to climate. While both are rapidly changing, these changes seem too vast and complex for everyday comprehension. Design Earth explores the aquarium as a device to enable public proximity and to develop a new mythology capable of reinstalling an aesthetic sense of wonder to the ocean world.

DESIGN EARTH is led by Rania Ghosn and El Hadi Jazairy (MIT; University of Michigan). Recipients of the Architectural League Prize for Young Architects and Designers, their work has also been supported by the Jacques Rougerie Foundation and Boghossian Foundation and it has been collected by the Museum of Modern Art. DESIGN EARTH has exhibited and published widely, at, for example, Venice Architecture Biennale (2018 and 2016); Seoul Biennale; Boston Design Biennale; Oslo Architecture Triennale; Times Museum, Guangzhou; and the Sursock Museum, Beirut. Ghosn and Jazairy are authors of *Geostories* (2018) and *Geographies of Trash* (2016) and editors of *New Geographies 2: Landscapes of Energy* (2009) and *New Geographies 4: Scales of the Earth* (2011).

*This essay was first published as "Leviathan in the Aquarium," *Journal of Architectural Education* 71, no. 2 (2017): 271–79. Reprinted with permission.

How do we make sense of the environment at a moment when human beings have declared themselves “geographic leviathans?” To live in an epoch that is shaped by extensive environmental transformations is to be confronted with uncertainties at the scale of the planet. Paradoxically, we remain so little mobilized in part because of our failures to represent the scales of a story that is difficult both to tell and to hear. The Pacific Aquarium project appropriates the object of the aquarium to take aim at the scalar dissonance between our selfish economic worries and the expansive scales of Earth.⁽¹⁾ The project constructs the future of the ocean in relation to anthropogenic dynamics—of gyres of marine debris, deep-sea mining plumes, and ocean acidification. Thus, the miniature channels our cultural enchantment with natural history and its propensity to make universal knowledge a personal experience and attempts to reclaim environmental externalities as an intimate part of the political constituency of Earth.

Although I sometimes participate in these critiques of ideologies of nature, I am increasingly convinced that for all the merit of our deconstructive erudition, we are left with a rather antiseptic nature which has little if any political appeal. [...] Where the analytical practicality of left critiques is at all implemented, it is largely subordinated to a science-driven, policy-oriented, bureaucracy centered project of environmental management. [...] I want to insist that the re-enchantment of nature... be mobilized against establishment environmentalism.

Neil Smith, “Nature at the Millennium: Production and Re-enchantment”

There will be unexpected consequences. Taking care of unexpected country will be required—again and always. Reconciliation is not guaranteed; it is proffered, suggested, haltingly pictured. Any reconciliation will depend on descendants of settler worlds letting go of salvation history and instead learning to live in technoculture in something more like the time of Aboriginal country, facing ancestors of many kinds and responsible for those who come after. Technocultural people must study how to live in actual places, cultivate practices of care, and risk ongoing face-to-face encounters with unexpected partners. Donna Haraway, “Speculative Fabulations for Technoculture’s Generations: Taking Care of Unexpected Country”

In 1853, Philip Henry Gosse, a self-taught naturalist and popularizer of natural science, created the first public aquarium at the London Zoo, and published *The Aquarium, An Unveiling of the Wonders of the Deep Sea*, the first manual on how to build and take care of a miniature ocean. An aquarium, he asserted, was “the perfect way to get acquainted with the peculiar creature of the ocean without having

to descend into the depths.”⁽²⁾ Until mid-century, the bottom of what the sailors call “blue water” was “so unknown to us as is the interior of any of the planets of our system.”⁽³⁾ The first systematic efforts to sound the open sea date from the mid-nineteenth century. The growth of commercial and strategic opportunities, such as transoceanic shipping, whaling, and submarine telegraphy, attracted investigators to the ocean’s third dimension. The aquarium genre quickly grew into a craze, first within the home then as a collective experience, comparable to funfairs, zoos, and natural history museums, which offered viewers the chance to contemplate aquatic creatures in miniatures of their environments, both for scientific study and popular pleasure.

Among the many early handbooks on aquarium care is *Ocean Gardens* by the British artist and naturalist Noel Humphreys. In his opening chapter, Humphreys admonished his reader: “To appreciate Nature, the mind requires a special education, without which the eye and the ear perceive but little of the miracles passing before them.” He added, “the wonders of the ocean floor do not reveal themselves to vulgar eyes.”⁽⁴⁾ The aquarium was born out of such a desire to represent to the senses, or make sense of, the inaccessible, expansive, and mysterious deep sea. As the locus of an unprecedented form of knowledge production, the aquarium made visible a new site of knowledge. Here, scientific, socio-economic, technological, and political forces were manifested, entailing new modes and aesthetics for the production of evidence.”⁽⁵⁾ It is both an aquatic cabinet of curiosities that surveys the terra incognita of the submarine world, and a miniature ocean, which by framing a portion of the underwater world gestures toward an unframeable whole. Wild speculations and fantasies are subsequently woven around the unfathomable abyss and “compressed into an easily comprehended menagerie, an oceanic garden in miniature, a submarine chamber of wonders.”⁽⁶⁾

Today, Humphreys’ call to develop our “vulgar eyes” is very timely as humans make sense of the ocean at a moment in which they have declared themselves “geographic leviathans.” The ocean is increasingly present in visualizations of Earth and of scientific accounts of the impact of industrial activities on the planet. Although the Blue Marble was a proto-environmentalist icon for a fragile ocean planet, for a few decades after the first space satellite images, the ocean remained a giant blank blue surface in web-based models of the planet. “We had this arbitrary distinction that if it was below sea level it didn’t count,” recalled John Hanke, the Internet entrepreneur who co-created the progenitor of Google Earth.⁽⁷⁾ Until 2009, all of the existing features on Google Earth—mountains, valleys, cities, plains, ice sheets—were built through programming from an elevation of zero up. It was only in 2009 that representations of submarine topography and other ocean data began to figure in Google Earth, and even then only 5 percent of the ocean floor is mapped in

detail.⁽⁸⁾ The ocean is also an inseparable dimension of geographies of climate change. About 93 percent of the excess heat energy stored by Earth over the last fifty years is found in the ocean. A major resource for life on Earth, the ocean is undergoing significant anthropogenic transformations—of gyres of marine debris, deep-sea mining plumes, and ocean acidification. In 2014, the Intergovernmental Panel on Climate Change (IPCC), which was set up in 1988 to inform policymakers of the science of climate change, gathered all the scientific knowledge for the first time in a single document, which extended interest in the ocean beyond the surface concern of rising sea levels to engage the totality of the role of the ocean in the climate system and climate change impacts.⁽⁹⁾ Several barriers impede the ability of the general public to think about and feel the impacts of deep-sea mining and climate change on the ocean. In general, complex systems are difficult to fathom, and the issue of climate change is particularly challenging because its effect involves vast scales of both time and space. While weather is both temporally and geographically closer and part of everyday life and experience, climate refers to patterns over time. Furthermore, an ocean is large, difficult to see all at once in its depths and extents, and requires a vast representational machine to comprehend. Its sheer size is overwhelming.

What is the affective agency of a cabinet of natural history in a post-natural world and within this ocean of uncertainties? If environmental issues are un-representable in their scale, their ubiquity, and their duration, then perhaps miniatures of Earth can present such scientific concerns to the senses. Miniaturization becomes a demonstration of the rule of containment: the microcosm encapsulates the scales of Earth, it is a *tableaux vivant* predicated on manipulation of crafted “props” in carefully defined technologies and landscapes. The reverie of the miniature is a pervasive feature of the human imagination, as Gaston Bachelard has shown in his explorations of the rich textures of the real. “Miniature is an exercise that has metaphysical freshness,” notes Bachelard. “It allows us to be world conscious at slight risk,” he adds, “and how restful this exercise on a dominated world can be! For miniature rests us without ever putting us to sleep. Here the imagination is both vigilant and content.”⁽¹⁰⁾ The microcosm renders the ocean visible, comprehensible and hence a possible domain of reflection and action. Beyond a symbolic surrogate of the totality, the microcosm is also world-building, which by channeling wonder and marvel, aspires to build a world beyond the strict values of the economy. Susan Stewart recognized such capacity of miniature objects. “The reduction in scale which the miniature presents skews the time and space relations of the everyday lifeworld, and as an object consumed the miniature finds its ‘use value’ transformed into the infinite time of reverie.”⁽¹¹⁾ Within such a complex web of environmental relationships between facts, uncertainties, emotions, and actions, the aquarium brings remote

scales, uncertain futures, and intangible science concepts, such as climate change, to the personal realm.

The Pacific Aquarium project was developed for and first exhibited at the Oslo Architecture Triennale “After Belonging.”⁽¹²⁾ The invitation to contribute to the section on “Markets and Territories of the Global Home” brought us to speculate on what it is to belong to Earth, and in particular to the Clarion-Clipperton Zone (CCZ) in the Pacific Ocean. The site of CCZ covers an area approximately the size of Europe and has the world’s largest deposits of deep-seabed rare earth minerals, which are used in the production of batteries and alloys. The seabed outside of national jurisdictions, called the “Area” in the United Nations Convention on the Law of the Sea (UNCLOS), is legally part of the “common heritage of mankind,” which following UN General Assembly Resolution 2749, should not be subject to national appropriation. The “Area” is under the legal mandate of the International Seabed Authority (ISA) that was created in 1982 with the mission to oversee exploration, mapping, and resource management in the High Seas. Since 2001 however, ISA has granted twelve exploration licenses for minerals on the CCZ deep seabed. The magnitude of the damage to the CCZ reserved for resource extraction cannot be fully anticipated, as the scale of mining is unprecedented and the ecosystems at this depth of the ocean are yet to be fully understood. Researchers have stated that “efforts to adapt to and/or mitigate the effects of climate change on marine ecosystems must operate in a framework characterized by uncertainty from multiple sources,” and that “the scientific community has to rely on scenarios to consider complexity and irreducible uncertainty.”⁽¹³⁾

Pacific Aquarium appropriates the object of the aquarium to take aim at the scalar distance between our selfish economic worries and the great scales of Earth. A series of nine aquariums, each addressing a particular economy-environment nexus, reclaim ocean space as public controversies by connecting political ecology with speculative design and a collective aesthetic experience. The Aquarium reckons as such with the cognitive and affective dissonance between what feels like minute individual concern and agency and a context of enormous collective impact. It also presents the following significant attributes:

1. The aquarium is a three-dimensional territory that materializes the vertical element of volume.⁽¹⁴⁾ It is inherently sectional.

2. The aquarium is populated, a multi-species space. It counters images of the sea that present it as a “territory of emptiness,”⁽¹⁵⁾ such as Lewis Carroll’s evacuated *Ocean Chart* from *The Hunting of the Snark*.⁽¹⁶⁾

3. The aquarium is *now*. Rather than the vastly accelerated aesthetics of contemporary Anthropocene representations, of

exponential info-diagrams or disaster satellite images, the aquarium favors an immersive, enchanting experience that speaks directly to climate change. The aquarium slows down thought and grounds it in the immediacy and wonder of the present.

4. The aquarium is also *there*. It is both a model of the world and an object of wonder. It actually capitalizes on its distance from the real through its transparent, yet clearly defined, boundary condition. Its technologies of the spectacle draw on the perceptual logics and plots of the shop window, the theater, the panorama, and literal and fictive travels to become an amalgam of all.⁽¹⁷⁾ With monsters, storms, a modern-day Noah, scientists, and jellyfish on stage, the aquarium is the setting in which the non-speaking and non-human entities operate as devices of estrangement. The aquarium's uncanny ability to draw on multiple, even competing, visual and textual logics is central to its work.

5. The aquarium holds vividly in tension the panorama and its details. It stages a slice of the ocean nesting a range of scales—from the micro to the macro—into one another, without ever fitting neatly.

The project drawings bring the attributes of the aquarium into a microcosm on a sheet of paper. They depict three-dimensional cross-sections of the ocean with a split view that shows the environments both above and below the water surface. This mode of representation, known popularly as the aquarium view, draws on illustrations in the biological sciences and geological sciences, such as the frontispiece from Charles Lyell's *Principles of Geology*, which depicts the rock cycle and Charles Darwin's illustration of coral reef formation,⁽¹⁸⁾ or *Duria Antiquior*, a pictorial representation of a scene of prehistoric life based on evidence from fossil reconstructions. The drawings also appeal to a geographical project that predates the disciplinary schism of sciences of the earth, and in particular to Alexander von Humboldt's "portrait of nature," which presented a totality of the scales and sciences of the earth in an assemblage of astronomy, geography, botany, and geology. Although Humboldt's avowed purpose was that of "scientific traveller," his drawings assembled forms of knowledge, which, beyond the accumulation of information, engaged the aesthetic experience of the reader. They made commensurability possible. "People want to see," Humboldt noted, "and I show them a microcosm on a sheet of paper."⁽¹⁹⁾

The aquarium view is also an act of fantastic world-building, much like the task of science fiction (or what Donna Haraway prefers to call "SF: Science Fiction, Speculative Fabulation, String Figures, So Far").⁽²⁰⁾ This model of the ocean resonates with Haraway's concept of "worlding" as a process of actively reimagining a non-anthropocentric world. "These knowledge-making and world-making fields," Haraway observes, "inform a craft that for me is relentlessly

replete with organic and inorganic critters and stories, in their thick material and narrative tissues.”⁽²¹⁾ The purpose of worlding is not to predict the future, let alone to fix it, but to raise questions about present relations between humans and the world they inhabit. The stories being told serve as such to explicate the here-and-now. They build on surrealism as the procedure of using violence to illustrate the present political-environmental-moral state of affairs.⁽²²⁾ “Sometimes the most direct way to tell the truth,” Ursula Le Guin reminds us, “is to tell a totally implausible story, like myth. That way you avoid the muddle of pretending the story ever happened, or ever will happen.”⁽²³⁾ The figures and events of myth also provide a medium through which universal environmental knowledge becomes personal experience and a common ground on which we can meet, not only rationally, but aesthetically and emotionally. Framed as such, their end is the assembly of a body that cares to speak to and with Earth.

Such fabulations are part of the process of developing a new mythology of the environment—a space for our fables, legends, and architectural precedents to be retold, rethought, and reenvisioned. Bruno Latour proposes that climate change calls for a new worldview that counters the objectification of Earth and its associated Promethean worldviews. In his view, the assumed divide between nature and society—and the accompanying focus on deanimate, disembodied, undisputed reason—has led directly to the current ecological crisis. We do not live on a “Blue Marble,” insofar as that famous image of our planet symbolizes an objective, holistic, impersonal earth made visible by our own technological achievements. Such metaphysics of technological progress, Latour argues, should now be countered by a redefined assemblage of *values*, so as to extend beyond the critique of the modern objectification of the Earth to a new ecological belief-system in the embodiment of Gaia. As its name evokes, the Greek goddess of Earth is an “odd, doubly composite figure... the Möbius strip of which we form both the inside and the outside, the truly global Globe that threatens us even as we threaten it.”⁽²⁴⁾

Each of the nine aquarium views below reckons with the uncertainties of environmental and technological futures in the Pacific Ocean.

CLASSIFIED SEDIMENTS [Fig. 1]

Nearly five centuries of marine discovery notwithstanding, the ocean remains largely unknown, unmeasured, and unmapped.⁽²⁵⁾ Those agencies and corporations that have invested the most in visualizing the CCZ are those that are only intent on gauging the value of the materials found in the seabed. The two-dimensional “floor plans” developed from their data clearly show the narrow scope of their interests. In Classified Sediments, an artificial landform camouflages the entrance to a sequence of vaults that extend deep

into Earth's crust to house security and intelligence records on the deep sea. This icon of secrecy dissimulates itself by mimicking the cartographic resolution of the trigonometric projection of the deep sea. The territory of the deep sea hides in the map.

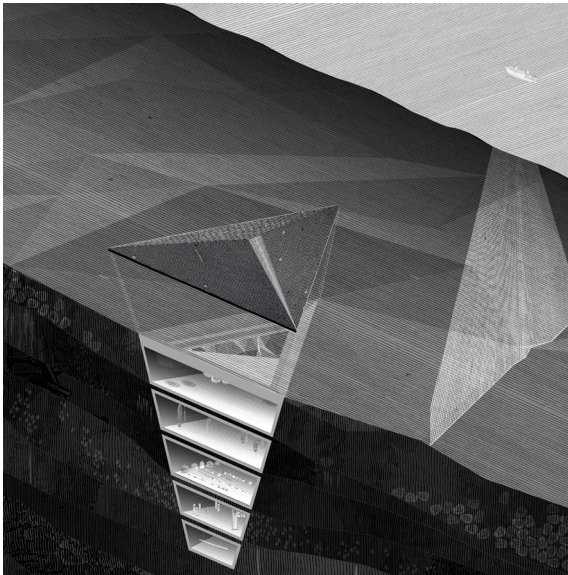
OVERMINING [Fig. 2]

With support from the ISA, a twenty-first-century gold rush is set to begin on the ocean floor, with likely consequences that may include changes to the geochemistry of the sediment and light and noise pollution from the machinery. The same ISA has also mandated a detailed procedure in order to conserve the flora and fauna of the mining area to facilitate an accelerated recovery of the ecosystem post-extraction. Each mining entity will be required to extract substrate samples along a geographically indexed system of transects on a 500 meter grid across its claim. Suspended by cables from a grid of surficial floats, the forest of linear striations is a Hanging Gardens of the Pacific, which similarly to its Babylonian namesake is subject of controversies on whether it is an actual construction or a poetic creation. Over a decade, the infrastructure incubates a benthic ecosystem that will be grafted onto the depleted seabed.

CYBORG FISH COLONY [Fig. 3]

The International Seabed Authority mandates the conservation of the flora and fauna in the mining area of the Clipperton fault. Deep-sea mining however produces plumes that smother near-bottom species away from their habitats. The project draws on the modernization of the fishing industry and the radical changes in capture techniques and in the fish themselves. With killing machines and robo-fishers ruling the oceans, the fish itself is transformed into

Fig. 1
Classified
Segments (Design
Earth, Pacific
Aquarium, 2016)



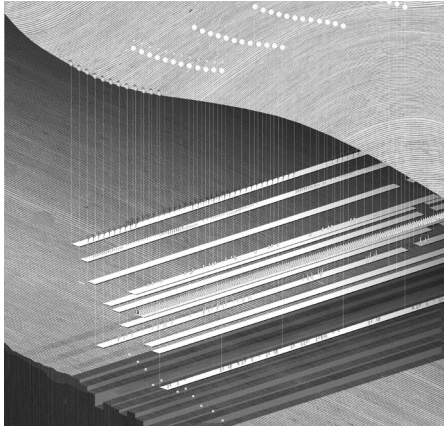


Fig. 2
Overmining
(Design Earth,
Pacific Aquarium,
2016)

a hybrid of machine and organism, a creature of social reality as well as a creature of fiction. A school of cyborg fish collects the plumes into a massive spherical sponge-like nest where the toxic particles are solidified and processed into energy. The fish subsequently decompose into organic sustenance for the returning species.

BELOW THE WATER TOWERS [Fig. 4, p. 274]

At the ocean floor the sharply toothed, combine-like suction machines will churn and rip the ocean surface into a sediment slurry, which if not contained would distribute widely through deep ocean currents. A catchment dome caps mining activities to contain sediment plumes. Polluted water is separated from surrounding water and transported into a series of inverted water towers just below the surface for processing. Purified water is gradually released back into the ocean. Under the Water Towers flips Manhattanism on its head. It displaces Buckminster Fuller's Dome over Manhattan, and rather than protecting the air of the city it processes the water of the deep sea.

MARINE LANDFILLS

Gyre plastic is matter in motion across the North Pacific Ocean connecting ocean ecosystems with cultures of consumption and waste.⁽²⁶⁾ It is estimated that without significant action, there may be more plastic than fish in the ocean, by weight, by 2050. Large-scale landfills capture floating waste, oil, fuel and detergents from the Pacific gyres. Seawater cascades into a landfill to be filtered by a one-way membrane surface. When a marine landfill site is filled, the inverted pyramid is sealed and becomes a floating island in the sea.

IRON TOWERS [Fig. 5, p. 274]

The uptake of carbon dioxide (CO₂) from the atmosphere is decreasing the pH of Earth's oceans. A number of ocean scientists and businesses are exploring iron fertilization as a means to stimulate photosynthesis in plankton, converting dissolved carbon dioxide

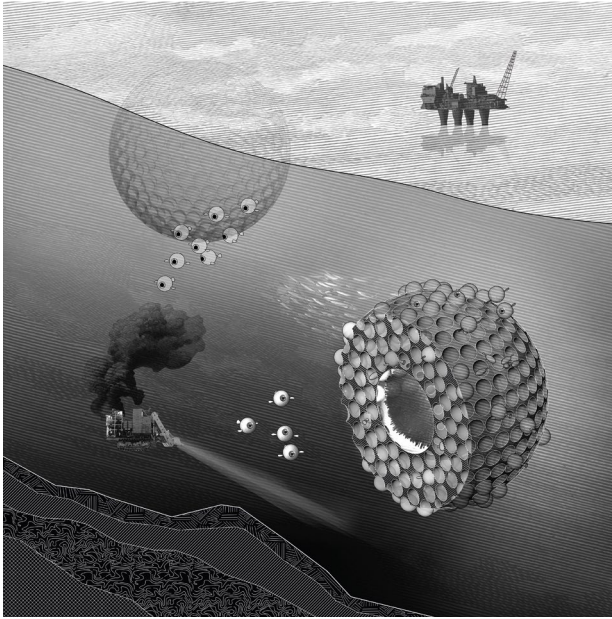


Fig. 3
Cyborg Fish Colony
(Design Earth,
Pacific Aquarium,
2016)

into carbohydrates and oxygen. The vertical tensile structure, which contains high iron concentrations of water in its tubes, extends the habitat of phytoplankton to the deep ocean attracting corals and other sea creatures to the infrastructure.

MEDUSA MAZE

Jellies, which bloom seemingly wherever ocean environments are destabilized, are a visual index of a dying ocean. As coral reefs, cetaceans, sea turtles, and fish die off, jellyfish will take stage as the main animal protagonists of oceans. As the seas become distressed. “the jellyfish are there, like an eagle to an injured lamb or golden staph to a postoperative patient—more than just symptoms of weakness, more like the angel of death.”⁽²⁷⁾ The Medusa Maze is a jellyfish Pac-Man, populated with planktons and sea turtles. The Maze is a jellyfish husbandry for an emerging cosmetic industry. It is also a *Damnatio ad bestias*,⁽²⁸⁾ an Anthropocene arena that pits the gelatinous beasts against their predators in a luminescent aquarium.

CLIMATE SANCTUARIES

A Small Island State (SIS) is a low-lying coastal country that is particularly vulnerable to sea level rise. When a SIS loses its land fully to the rising sea level, its defined territory ceases to exist, hence no longer meeting one of the four UN criteria for statehood.⁽²⁹⁾ The Climate Sanctuaries project reconstructs the territories of the first five submerged Pacific islands of Vanuatu, Marshalls, Fiji, Tuvalu, and Kiribati through their cultural landscapes: volcanic mountains, coastlines, highlands villages, cliffs, and water territories.

PARLIAMENT OF REFUGEES

The challenge of governing the ocean is that it lacks formal representation—both political and aesthetic—through which to think the new environments of climate change. The ocean hovers awkwardly outside the zone of national claims, like an uninvited guest without seats at the table. If politics, as Latour articulates, “is the art of cohabiting together in this immense greenhouse with species that have very different demands for survival,”⁽³⁰⁾ then what is the greenhouse, or should we say “blue-house” of climate change? What assemblies could stage a totality, especially when that whole is opaque, fragmented, contradictory?

The Parliament of Refugees contains within the same artificial envelope different things and species of the Anthropocene, such as sea turtles, CO₂ molecules, scallops, bleached corals, drowning wetlands, hammerhead sharks, algae, *Homo sapiens*, *Brighamia rockii*, Nihoa finch. The Assembly is organized around a hollow pillar that connects it to the center of the earth through a submerged volcano.

CONCLUSION

To live in an epoch shaped by such anthropogenic transformations is to be confronted with risks at the scale of the planet. Paradoxically, we remain so little mobilized in part because of our failures to address uncertainties and to represent the scales of a story that is difficult both to tell and to hear. The current environmental condition seems to involve a crisis of the imagination, the amendment of which might depend on finding other forms and forums of eco-political engagement. Pacific Aquarium discloses its epoch’s capacity to feel that there are other possible ways for the environment to consist.⁽³¹⁾ The miniature of Earth explores an alternative mode of environmental narration in order to reclaim and cultivate response-ability. In the midst of spiraling ecological devastation, this speculative fabulation is “not interested in reconciliation or restoration,” but is “deeply committed to the more modest possibilities of partial recuperation and getting on together. Call that staying with the trouble.”⁽³²⁾ For speculative fabulation, Haraway reminds, “It matters what thoughts think thoughts. It matters what knowledges know knowledges. It matters what relations relate relations. It matters what worlds world worlds. It matters what stories tell stories.”⁽³³⁾ Charged with wonder, such environmental fabulation might be the necessary storytelling for the geographic leviathan’s earthly survival. It channels enchantment to reclaim territorial externalities within the political constituency of Earth.

- (1) Pacific Aquarium Project Credits: Rania Ghosn and El Hadi Jazairy, with Reid Fellenbaum, Ya Suo, Jia Weng, Shuya Xu, Saswati, Rixt Woudstra, Alaa Quraishi, Namjoo Kim, Ammar Ahmed, Alexandra Chen.
- (2) Bernd Brunner, *The Ocean at Home: An Illustrated History of the Aquarium* (Princeton: Princeton Architectural Press, 2005), 41.
- (3) Quote by Matthew Fontaine Maury, in Helen M. Rozwadowski, *Fathoming the Ocean: The Discovery and Exploration of the Deep Sea* (Cambridge, MA: Harvard University Press, 2008), 33.
- (4) Noel Humphreys, *Ocean Gardens* (London: Sampson Low & Sons, 1857), 3-9.
- (5) Natascha Adamowsky, *The Mysterious Science of the Sea, 1775-1943* (London: Routledge, 2015), 103.
- (6) Brunner, *Ocean at Home*, 126.
- (7) Andrew C. Revkin, "Google Earth Fills its Watery Gaps," *New York Times* (February 2, 2009), <http://www.nytimes.com/2009/02/03/science/earth/03oceans.html>
- (8) Stefan Helmreich, "From Spaceship Earth to Google Ocean: Planetary Icons, Indexes, and Infrastructures," *Social Research* 78, No. 4 (2011): 1211-42.
- (9) IPCC, "Climate Change 2014: Synthesis Report," <https://archive.ipcc.ch/report/ar5/syr/>
- (10) Gaston Bachelard, *The Poetics of Space*, trans. Maria Jolas (Boston: Beacon Press, 1994 [1958]), 161.
- (11) Susan Stewart, *On Longing: Narratives of the Miniature, the Gigantic, the Souvenir, the Collection* (Durham, NC: Duke University Press, 1993), 65.
- (12) For further information about the Triennale, visit <http://www.afterbelonging.org>
- (13) M. R. Payne et al., "Uncertainties in projecting climate-change impacts in marine ecosystems," *ICES Journal of Marine Science* 73, no. 5 (2016): 1273,1278.
- (14) Philip Steinberg, "Wet Ontologies, Fluid Spaces: Giving Depth to Volume through Oceanic Thinking," *Environment and Planning D: Society and Space* 33, (2015): 247-64. For more on recent literature on the sea in geographical research, see Kimberley Peters, *Water Worlds: Human Geographies of the Ocean* (London: Routledge, 2016).
- (15) A term used by Alain Corbin to describe the seaside before its appropriation by nineteenth-century Europe. Alain Corbin, *Le Territoire du Vide: l'Occident et le Desire du Rivage 1750-1840* (Paris, 1988); *The Lure of the Sea: The Discovery of the Seaside in the Western World, 1750-1840*, trans. Jocelyn Phelps (Cambridge: Polity, 1994).
- (16) Lewis Carroll, *The Hunting of the Snark*, 1876.
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The book tells the story of the sea-land continuum based on the case of the North Sea—one of the world's most industrialised seas, in which the Netherlands plays a central role. The space of the North Sea is almost fully planned and has been loaded with the task of increased economic production from new and traditional maritime sectors. At the same time, it has been emptied of cultural significance.

Through diverse projects from academia, art, literature, and practice, from analysis to design, the book explores synergies for designing this new spatial realm. Port city expert Carola Hein, professor of the history of architecture & urban planning at Delft University of Technology, and Nancy Couling, associate professor at the Bergen School of Architecture and researcher of the urbanised sea, combine forces with interdisciplinary experts to guide the reader through this complex and fascinating topic.



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